

INTEGRATIVE ANALYSIS OF RNA SEQ AND CHIP SEQ DATA IN K562 & MCF7 CELL LINE

Next-generation sequencing (NGS) methodologies, such as RNA-seq and ChIP-seq, have revolutionized genomic research by providing unprecedented insights into gene expression, regulation, and epigenetics. In this paper, we review the successful applications of RNA-seq and ChIP-seq in cancer and genetic disease research, with a specific focus on the K562 and MCF7 cell lines as representative models of leukemia and breast cancer, respectively. These techniques enable the elucidation of key biological processes, identification of biomarkers, and discovery of therapeutic targets, ultimately advancing our understanding and treatment strategies for complex diseases.

The datasets used in this study were obtained from the ENCODE (Encyclopedia of DNA Elements) database, a comprehensive resource for genomic data. For the K562 cell line, RNA-seq data (ENCFF002DKA and ENCFF002DKE) and ChIP-seq data (ENCFF839LPL and ENCFF698ICA) were acquired. These datasets allow for the exploration of gene expression patterns and transcription factor binding sites associated with leukemia biology. Moreover, for the MCF7 cell line, RNA-seq data (ENCFF000HQR and ENCFF000HQP) and ChIP-seq data (ENCFF002EFD and ENCFF002EFA) were obtained. These datasets offer insights into gene expression dynamics and hormone receptor-mediated signaling pathways relevant to breast cancer biology.

By integrating and analyzing these diverse genomic datasets, researchers can gain a comprehensive understanding of the molecular mechanisms underlying leukemia and breast cancer. Furthermore, the identification of key genes, regulatory elements, and signaling pathways provides valuable insights into disease pathogenesis and potential therapeutic interventions. Overall, the utilization of RNA-seq and ChIP-seq data from the K562 and MCF7 cell lines offers a powerful approach to advancing our knowledge of cancer biology and improving patient outcomes.

METHODS AND MATERIALS

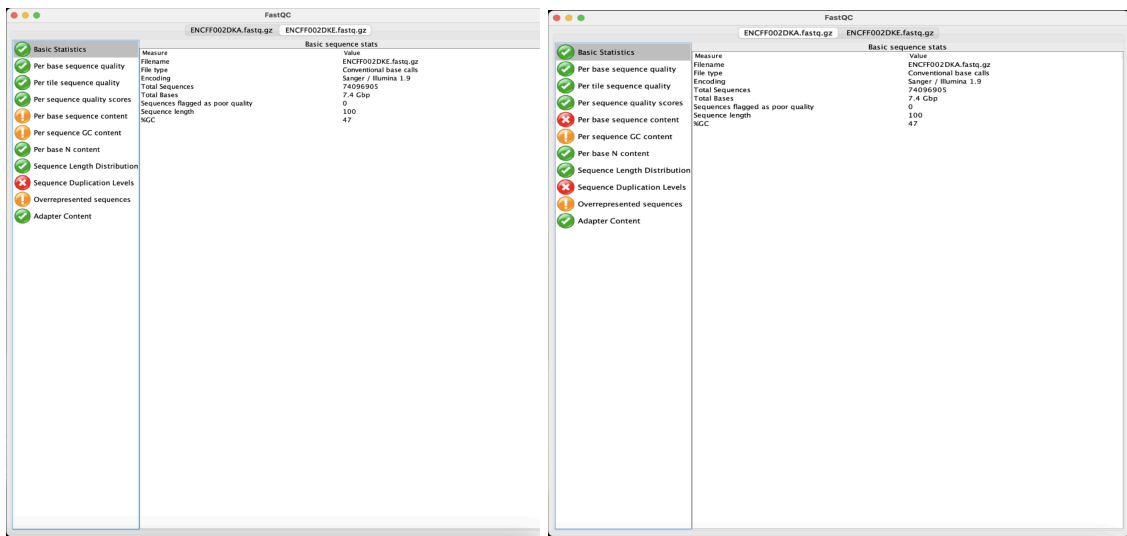
In our study, we employed a suite of bioinformatics tools to analyze RNA-seq and ChIP-seq data obtained from the K562 and MCF7 cell lines. **FastQC** was utilized for initial quality control assessments, providing crucial insights into the sequencing data's quality through metrics such as per-base sequence quality and GC content distribution. Subsequently, **Bowtie2** facilitated the alignment of sequencing reads to the reference genome, ensuring accurate mapping for downstream analysis. **Cufflinks** played a pivotal role in transcriptome assembly and differential expression analysis, enabling the estimation of transcript abundances and identification of differentially expressed genes. Furthermore, **DAVID** and **Cytoscape** were employed for functional annotation and network visualization, respectively, aiding in the interpretation of omics data and identification of key biological processes and regulatory elements. Additionally, **MACS2** and **MEME-ChIP** were utilized for peak calling and motif discovery in ChIP-seq data, uncovering transcription factor binding sites and regulatory motifs within genomic regions of interest. Additionally, the **Integrative Genomics Viewer (IGV)** software played a crucial role in our analysis, serving as a genome browser for the visualization and exploration of genomic data. Collectively, these tools provided a comprehensive framework for analyzing high-throughput

sequencing datasets, allowing us to gain valuable insights into gene expression, regulation, and epigenetics in the context of leukemia and breast cancer.

RESULTS

QUALITY CONTROL

We initiated our research by acquiring paired sequencing data (Reads1 and Reads2) from the ENCODE database. Employing FastQC, we conducted a thorough assessment of the quality of these files to confirm their precision and dependability. This crucial step enabled us to detect any potential issues that might impact our analysis, thereby ensuring that our research was founded on sequencing data of impeccable quality. Subsequently, we present the results obtained from the quality control analysis conducted using the FastQC software tool.



BOWTIE ALIGNMENT

After quality assessment, we used Bowtie2 to align paired-end sequencing reads to the human reference genome (hg19), ensuring precise mapping to genomic locations. Subsequently, Samtools facilitated post-alignment processing by converting SAM files to BAM format and performing sorting and indexing for efficient data manipulation. Additionally, BCFtools provided utilities for variant calling and genetic variation analysis. This streamlined approach forms a robust framework for analyzing genomic data, offering valuable insights into the functional genomics of K562 and MCF7 cell lines.

```
Last login: Mon Feb 12 19:58:24 on tty002
(base) shetty@Srvayas-MacBook-Air ~ % bowtie2 -x ./hg19/hg19 -1 ENCFF002DKA.fastq.gz -2 ENCFF002DKE.fastq.gz -S ENCFF002DKA_DKE_Aligned.sam
74096905 reads; of these:
  74096905 (100.00%) were paired; of these:
    18447072 (24.90%) aligned concordantly 0 times
    31206255 (42.12%) aligned concordantly exactly 1 time
    24443578 (32.99%) aligned concordantly >1 times
    -----
    18447072 pairs aligned concordantly 0 times; of these:
      4038037 (21.89%) aligned discordantly 1 time
    -----
    14409035 pairs aligned 0 times concordantly or discordantly; of these:
      28818070 mates make up the pairs; of these:
        18055514 (62.65%) aligned 0 times
        7068549 (24.53%) aligned exactly 1 time
        3694007 (12.82%) aligned >1 times
87.82% overall alignment rate
```

MUTATION CALLING AND VISUALIZATION

This process yields valuable insights into potential genetic changes underlying phenotypic differences or susceptibility to diseases. Utilizing aligned files (Reads1:

ENCFF002DKA, Reads2: ENCFF002DKE) from library ENCLB822JYE, mutation calling was conducted using bcftools. The human hg19 reference genome (hg19.fa) served as the template for variant detection. Basic variant statistics were derived from bcftools analysis to comprehend the mutation landscape. Additionally, visualization and exploration of specific genomic regions were facilitated by the Integrated Genome Viewer (IGV) software.



VISUALIZATION OF HBD REGION

CALCULATING GENE EXPRESSIONS

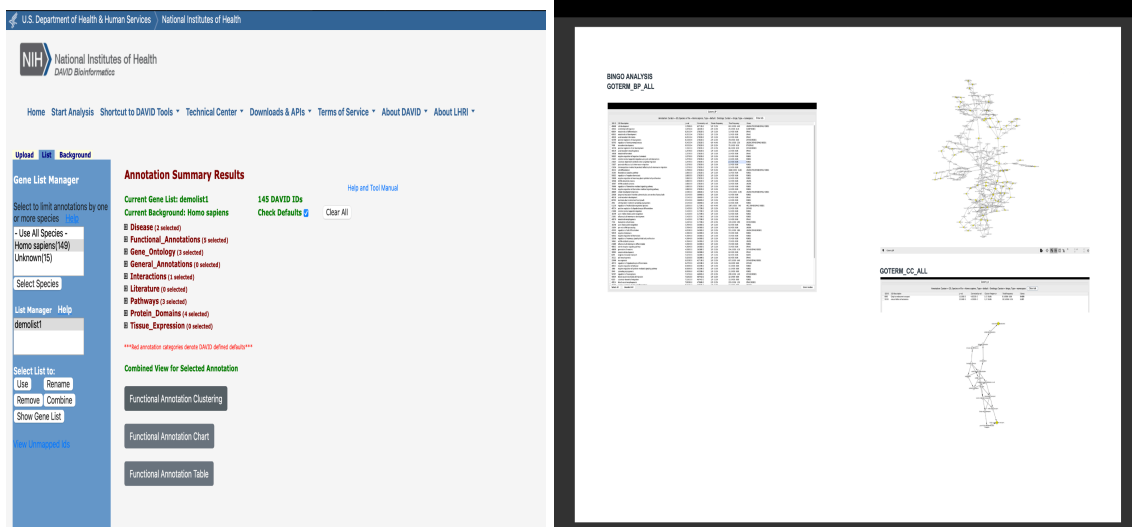
We began our analysis by acquiring RNA-seq datasets for the K562 and MCF7 cell lines from the ENCODE project, obtaining Reads1 and Reads2 for each cell line using specific Accession IDs. These reads were then aligned to the hg19 reference genome using Bowtie2, ensuring precise mapping to genomic coordinates. Following alignment, samtools was employed to filter uniquely mapped reads, ensuring the quality of alignments. Using Cufflinks, we quantified gene expression levels, providing insights into transcript abundance and isoform diversity. Differential expression analysis between the K562 and MCF7 cell lines was performed using Cuffdiff, allowing us to identify genes exhibiting significant expression differences.

In our study, we employed Cufflinks software to conduct gene expression analysis using the hg19_ncbiRefSeq_annotation.gtf file and the ENCFF800HQR_HQP_UniquelyAligned_Sorted.bam file. The software efficiently loaded the reference annotation and proceeded to inspect reads to determine the fragment length distribution. A total of 34,888 loci were processed during the analysis, revealing a normalized map mass of 68,499,429 and a raw map mass of 68,489,429 fragment length distribution. The empirical (learned) fragment length distribution estimated a mean of 277.42 and a standard deviation of 34.27. Transcript abundances were estimated as part of the analysis process.

```
* ~ cufflinks -G hg19.ncbiRefSeq_annotation.gtf ENCF000HQR_HQP.UniquelyAligned.Sorted.bam -o ENCF000HQR_HQP.GeneExpression
Warning: Could not connect to update server to verify current version. Please check at the Cufflinks website (http://cufflinks.cbc.b.umd.edu).
[04:25:25] Loading reference annotation.
[04:25:34] Inspecting reads and determining fragment length distribution.
> Processed 34880 loci. [*****] 100%
> Map Properties:
> Normalized Map Mass: 68409429.00
> Raw Map Mass: 68409429.00
> Fragment Length Distribution: Empirical (learned)
> Estimated Mean: 277.42
> Estimated Std Dev: 34.27
[04:31:02] Estimating transcript abundances.
> Processed 34880 loci. [*****] 100%
```

Following the execution of this command, Cuffdiff will conduct statistical analysis to detect genes and transcripts exhibiting differential expression between the two cell lines (K562 and MCF7). This analysis relies on the alignment files and annotation file provided as input.

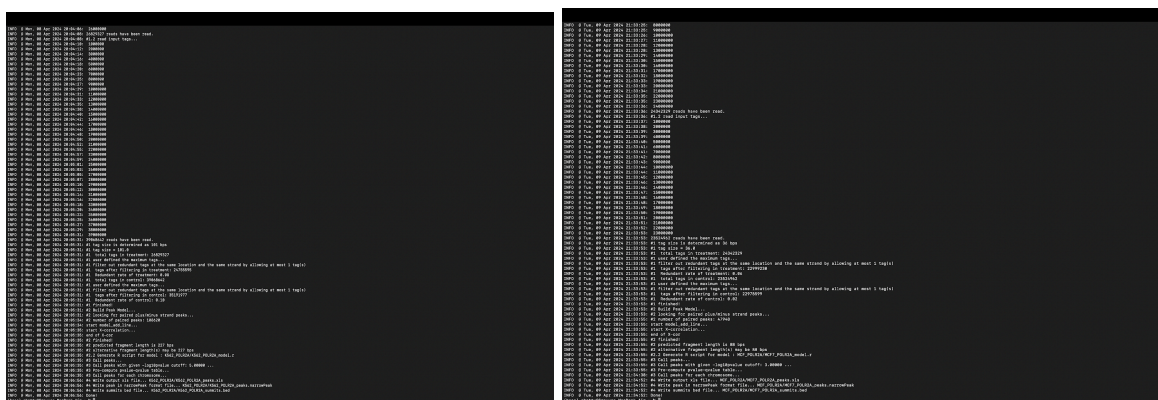
Sample 1		Sample 2		Gene		Transcript		Sample 1		Sample 2		Status		Value 1		Value 2		Log2(Fold Change)		Test Stat		P Value		Q Value		Significance	
test_id	gene_id	transcript_id	transcript_name	transcript_type	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status	transcript_status
* yes																											
431491	AASS	AASS	AASS	transcript	transcript	transcript	transcript	OK	0.0134722	0.72478	0.73109	-3.85413	0.00011814	0.0113714	yes												
431492	ABAT	ABAT	ABAT	transcript	transcript	transcript	transcript	OK	0.0118445	28.2011	11.1258	-4.16184	0.00000000	0.00000000	yes												
431493	ABCA1	ABCA1	ABCA1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431494	ABHD17C	ABHD17C	ABHD17C	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431495	ABRACL	ABRACL	ABRACL	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431496	ACVR2	ACVR2	ACVR2	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431497	ACMTF2	ACMTF2	ACMTF2	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431498	ADCV9	ADCV9	ADCV9	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431499	ADGRG3	ADGRG3	ADGRG3	transcript	transcript	transcript	transcript	OK	0.0110388	0.14388	0.00142	-4.28177	0.00000000	0.00000000	yes												
431500	ADGRG3	ADGRG3	ADGRG3	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
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431506	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.0100000	0.0000000	0.0000000	-0.0000000	0.00000000	0.00000000	yes												
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431508	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431509	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
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431513	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
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431516	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431517	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431518	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431519	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431520	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431521	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431522	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431523	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431524	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431525	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431526	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431527	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431528	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431529	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
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431535	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431536	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
431537	ANKRD1	ANKRD1	ANKRD1	transcript	transcript	transcript	transcript	OK	0.00000000	0.00000000	0.00000000	-0.00000000	0.00000000	0.00000000	yes												
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ANALYZING ChIP-SEQ DATA

Raw FASTA files for POLR2A ChIP-seq data from K562 and MCF7 cell lines were acquired from the ENCODE project. Using MACS2, a widely adopted tool for identifying transcription factor binding sites, we conducted peak calling to align sequencing reads and pinpoint POLR2A binding peaks across the genome. Through this analysis, we gleaned valuable insights into the regulatory landscape of transcription in both cell lines.

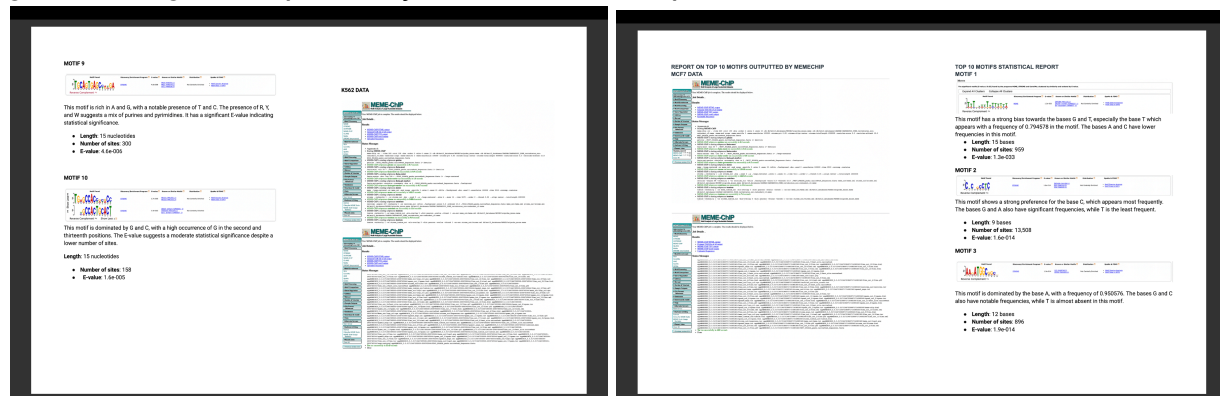
The analysis employed specific parameters, including a stringent P-value cutoff of $1.00e-05$, a bandwidth of 300, and an effective genome size of $2.70e+09$ base pairs, ensuring robust peak calling for POLR2A binding sites. Quality assessment revealed high-quality datasets, with minimal redundancy rates and consistent tag sizes of 101 base pairs across samples. The estimated fragment length of 227 base pairs provides crucial information for accurate peak identification. Overall, the MACS ChIP-seq analysis successfully identified significant POLR2A binding sites in both K562 and MCF7 cell lines, offering insights into its regulatory roles and genomic interactions for further investigation.



FINDING MOTIFS FROM ChIP-SEQ DATA

The analysis successfully uncovered significant DNA motifs within the POLR2A peaks identified in both the K562 and MCF7 cell lines. Through the utilization of MEME-ChIP, sequences corresponding to the narrow peaks were isolated and subjected to motif

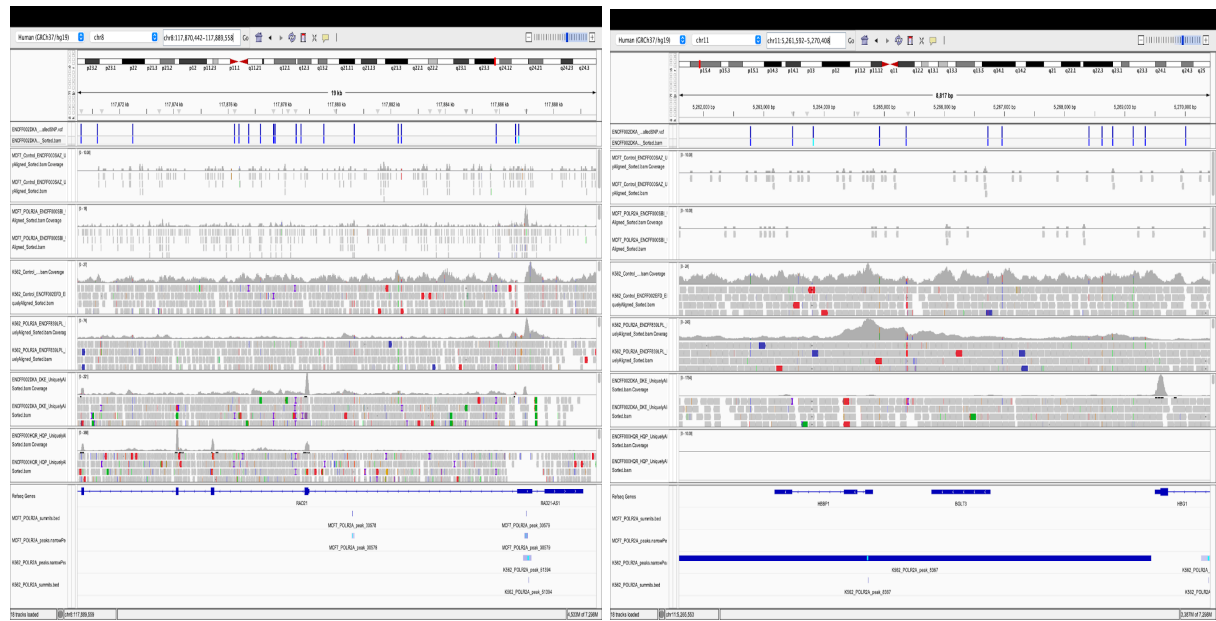
analysis. By referencing the JASPAR Vertebrate and UniProbe Mouse databases, enriched motifs indicative of transcription factor binding sites were discerned within the genomic regions captured by POLR2A ChIP-seq.



The MEME and STREME analyses revealed a series of conserved motifs within the provided sequences, ranging from 10 to 15 nucleotides in length. These motifs displayed varied nucleotide compositions, with preferences observed for certain nucleotides across different motifs. Notably, all identified motifs exhibited extremely low E-values, highlighting their statistical significance and suggesting that they are unlikely to have arisen by chance. Such high significance underscores the potential biological relevance of these motifs. For instance, the nucleotide compositions of these motifs could indicate potential binding sites for transcription factors or other regulatory proteins. Moreover, the specific nucleotide preferences within these motifs may offer clues about their roles and functions within the genome. Overall, these findings provide valuable insights into potential regulatory elements or functional sequences, emphasizing the need for further experimental validation to elucidate their exact biological roles and mechanisms of action.

IGV VISUALIZATION

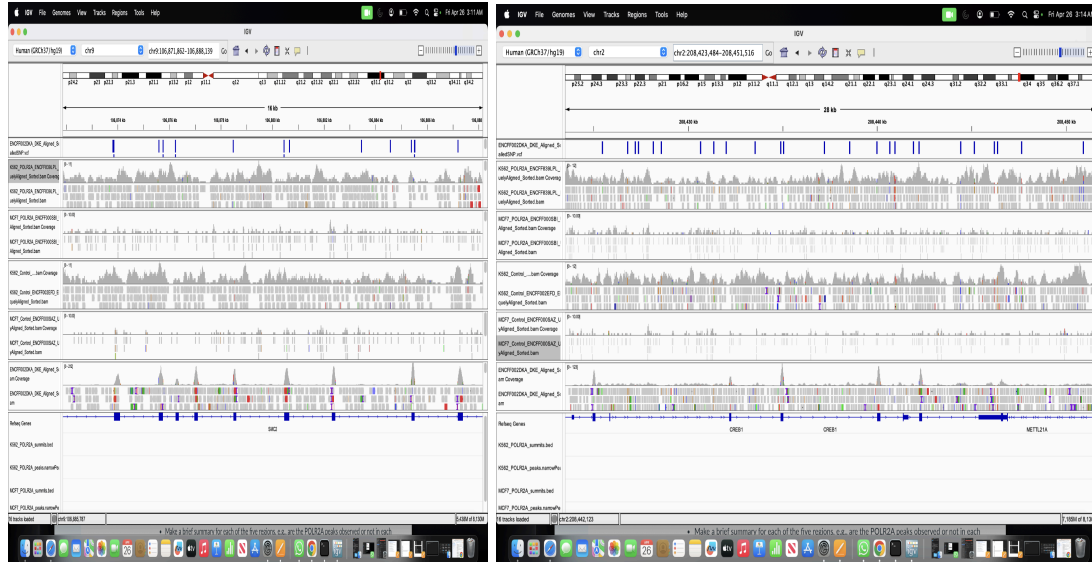
This segment delves into integrating and visualizing comprehensive data from both K562 and MCF7 cell lines using the Integrative Genomics Viewer (IGV), utilizing the hg19 genome version for customized tracks. The data upload includes uniquely mapped RNA-seq and POLR2A ChIP-seq reads, along with POLR2A peak positions and RNA-seq mutations for thorough analysis. We will focus on five specific chromosomal regions of interest, including chr11, chr8, chr9, chr17, and chr2, which encompass genes such as HBG1, RAD21, SMC2, POLR2A, and CREB1. Through this integrated approach, we aim to gain deeper insights into the genomic landscapes of K562 and MCF7 cell lines.



chr8:117840000-117920000 (RAD21 gene region) chr11:5250000-5282000 (HBG1 gene region)



chr17:7380000-7430000 (POLR2A gene region)



chr9:106840000-106920000 (SMC2 gene region) chr2:208390000-208485000 (CREB1 gene region)

CONCLUSION

In conclusion, our integrative analysis of RNA-seq and ChIP-seq data in the K562 and MCF7 cell lines has provided valuable insights into the molecular mechanisms underlying leukemia and breast cancer. Through the utilization of next-generation sequencing methodologies, we have elucidated gene expression patterns, regulatory elements, and epigenetic modifications associated with these complex diseases. Our study highlights the successful applications of RNA-seq and ChIP-seq in cancer research, demonstrating their potential in identifying biomarkers, therapeutic targets, and signaling pathways relevant to disease pathogenesis.

By leveraging datasets from the ENCODE database and employing a suite of bioinformatics tools, we have performed comprehensive analyses, including quality control, alignment, transcriptome assembly, differential expression analysis, functional annotation, peak calling, motif discovery, and genome visualization. These analyses have enabled us to uncover key biological processes, regulatory networks, and transcription factor binding sites implicated in leukemia and breast cancer biology.

Furthermore, our findings underscore the importance of integrative approaches in genomic research, where the combination of multiple high-throughput sequencing techniques allows for a more holistic understanding of complex biological systems. The identification of differentially expressed genes, enriched functional terms, and significant DNA motifs provides a foundation for further experimental validation and mechanistic studies.

Overall, our study contributes to advancing our knowledge of cancer biology and offers potential avenues for the development of novel diagnostic tools and therapeutic interventions. By integrating diverse genomic datasets and employing cutting-edge

analytical methods, we aim to facilitate the translation of research findings into clinical applications, ultimately improving patient outcomes in the fight against cancer.

REFERENCES

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